

Talk @ Azim Premji University

Statistics and Machine Learning as key technologies for society

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Why this Talk?



Let data drive decisions, not the **Highest Paid Person's Opinion**.

“Data is the new oil. It’s valuable, but if unrefined, it cannot really be used. It has to be changed into gas, plastic, chemicals, etc. to create a valuable entity that drives profitable activity; so must data be broken down and analyzed for it to have value.”

- Clive Humby, UK Mathematician and Architect of Tesco’s Clubcard.

Example:

City	Morning temperature	Evening temperature
Austin	62	90.7
Boston	41	48.0
Chicago	51	57.2
Denver	45	52.5

THE INTERNET IN 2023 EVERY MINUTE



"It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts."

- Sherlock Holmes, in A Scandal in Bohemia.



Astronomy



Social Networks



Healthcare



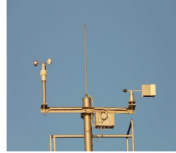
Banking



Genomics



Weather
measurements



- 1 **Statistics** is the study of the collection, analysis, interpretation, presentation, and organization of data.
- 2 **Data science** is the study of the generalizable extraction of knowledge from data, yet the keyword is science.
- 3 **Machine learning** is the sub-field of computer science that gives computers the ability to learn without being explicitly programmed.
- 4 **Artificial Intelligence** research is defined as the study of intelligent agents: any device that perceives its environment and takes actions that maximize its chance of success at some goal.
- 5 **Big data** is an all-encompassing term for any collection of data sets so large and complex that it becomes difficult to process using traditional data processing applications.

TOPIC 1 : STATISTICS

“Statistics is the universal tool of inductive inference, research in natural and social sciences, and technological applications. Statistics must have a clearly defined purpose, one aspect of which is scientific advance and the other, human welfare and national development”

- Professor P C Mahalanobis.

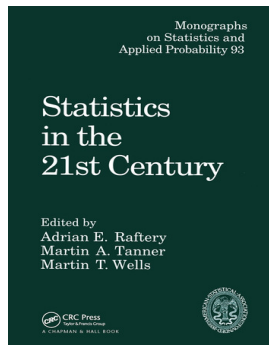
- **Role of Statistics:**

- 1 Making inference from samples
- 2 Development of new methods for complex data sets
- 3 Quantification of uncertainty and variability

- **Two Views of Statistics:**

- 1 Statistics as a Mathematical Science
- 2 Statistics as a Data Science

- Parametric Models : One Sample, two sample, linear models, survival data, Estimation, Testing of Hypothesis.
- Probability distributions were believed to generate data (e.g., Gaussian, Logistic, Poisson, Exponential, etc.).
- Semiparametric & Nonparametric Models : Dropping assumptions on population, dependence and errors.
- Emphases on Optimality in various ways : Bayes optimality, Decision theory, minimax and unbiasedness.
- Exact distributional (t, F) approaches and asymptotic methods (samples size $\rightarrow \infty$ viewed as approximation).



- Data : Large bodies of data with complex data structures are generated from computers, sensors, manufacturing industries, etc.
- Models : Non/Semiparametric models but in complex probability spaces / high-dimensional functional spaces (e.g., deep neural net, reinforcement learning, decision trees, etc.).
- Emphases : Making predictions, causation, algorithmic convergence.
- **Data** are necessary and at the core of Statistical Learning, Data Science & Machine Learning.
- **Statistics** : Not only has strong interactions with Probability but also other parts of Data Science (Machine Learning, Artificial Intelligence, etc.).

- **Probability** : Has moved to the center of Mathematics and having strong interactions with Statistical Physics and Theoretical Computer Science.
- **Statistics** : Not only has strong interactions with Probability but also other parts of Data Science (Machine Learning, Artificial Intelligence, etc.).
- **Computational** : Computing skills are essential, construction of fast training algorithms and computation time.
- **Applications** : Strong interactions with substantive fields in all areas. Applications of statistical methods in almost all the fields are evident. Statistics became a key technology driven by data ("**Data is the new oil**").



- **Traditional Problems in Applied Statistics:**
 - Well formulated question that we would like to answer.
 - Expensive to gather data and/or expensive to do computation.
 - Create specially designed experiments to collect high quality data.
- **Current Situation :** Information Revolution
 - Improvements in computers and data storage devices.
 - Powerful data capturing devices.
 - Lots of data with potentially valuable information available.

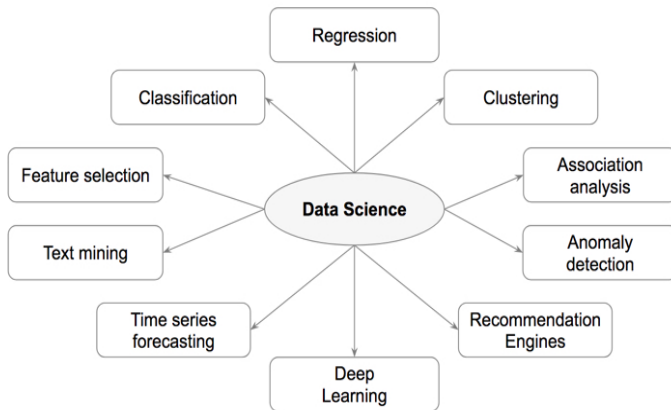
- Data characteristics:
 - Size
 - Dimensionality
 - Complexity
 - Messy
 - Secondary sources
- Focus on generalization performance :
 - Prediction on new data
 - Action in new circumstances
 - Complex models needed for good generalization
- Computational considerations :
 - Large scale and complex systems

TOPIC 2 : DATA SCIENCE

What is Data Science?

- *Interdisciplinary field about scientific methods, processes, and systems to extract knowledge or insights from complex data in various forms, either structured or unstructured.*
- *A concept to unify Statistics, Data Analysis and their related methods in order to “understand and analyze actual phenomena” with data.*
- *Employs techniques and theories drawn from many fields within the broad areas of Mathematics, Statistics, Information Science, and Computer Science, in particular from the subdomains of Machine learning, classification, cluster analysis, data mining, databases, and visualization.*
- *Fourth paradigm of Science (empirical, theoretical, computational and data-driven)*

Types of Data Science?



Basic Definitions:

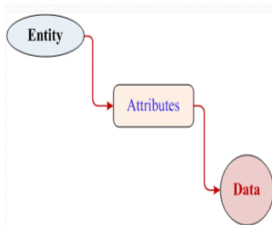
Entity: A particular thing is called entity or object.

Attribute: An attribute is a measurable or observable property of an entity.

Data: A measurement of an attribute is called data.

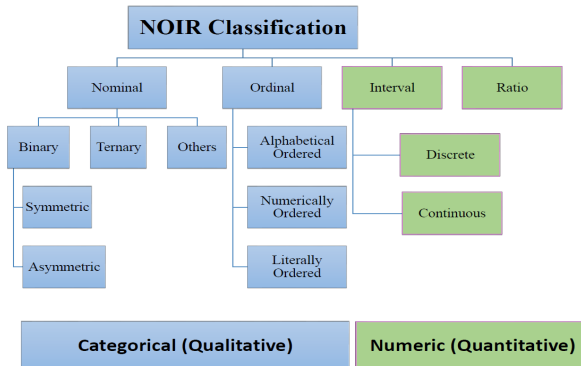
Note: Data defines an entity and Computer can manage all type of data (e.g., audio, video, text, etc.). In general, there are many types of data that can be used to measure the properties of an entity.

Scale: A good understanding of data scales (also called scales of measurement) is important. Depending on the scales of measurement, different techniques are followed to derive hitherto unknown knowledge in the form of patterns, associations, anomalies or similarities from a volume of data.



NAME	AGE	GENDER	SALARY	EMPLOYER
:				
:				
ABCD	34	F	40000	XYZ
:				
:				

- The **NOIR scale** is the fundamental building block on which the extended data types are built.
- Further, nominal (Blood groups, Attendance) and ordinal (Shirt size) are collectively referred to as **categorical or qualitative data**. Whereas, interval (weight, temperature) and ratio (Sound intensity in Decibel) data are collectively referred to as **quantitative or numeric data**.



Concept of data cube:

A **multidimensional data** model views data in the form of a cube. A data cube is characterized with two things:

- **Dimension:** The perspective or entities with respect to which an organization wants to keep record.
- **Fact:** The actual values in the record.

Example: Rainfall data of Meteorological Department.

- Time (Year, Season, Month, Week, Day, etc.)
- Location (Country, Region, State, etc.)

- In this 2-D representation, the rainfall for “North-East” region are shown with respect to different months for a period of years...

Region: North-East

	Month											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year	2005											
	2006											
	2007											
	2008											
	2009											
	2010											

- Suppose, we want to represent data according to times (Year, Month) as well as regions of a country say East, West, North, North-East, etc.

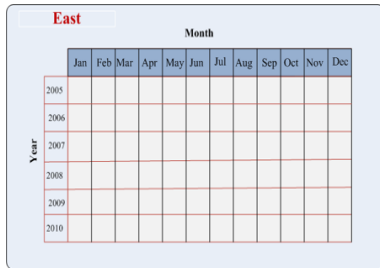


Figure: 2-D view of rainfall data

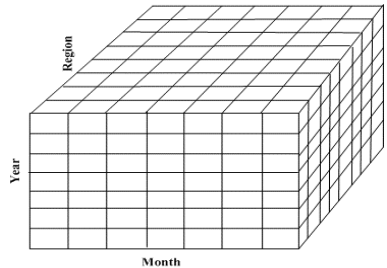
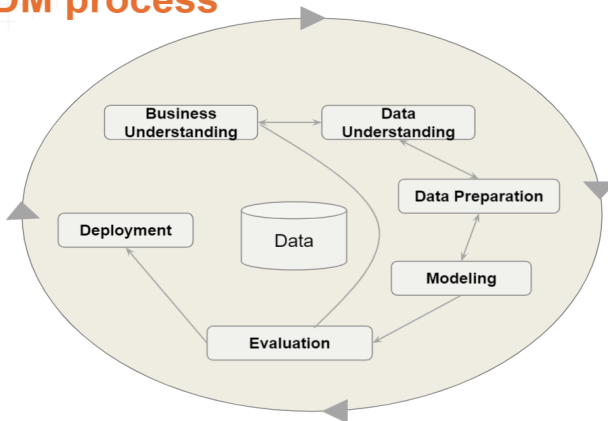
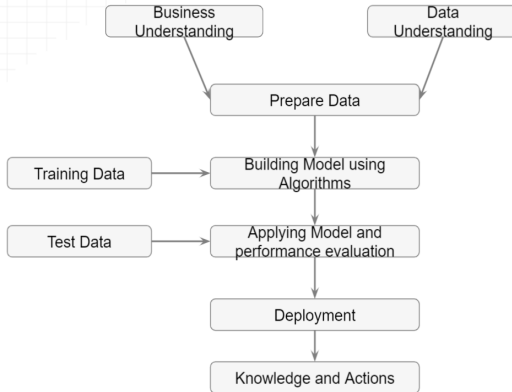


Figure: 3-D view of rainfall data

DM process



Process



1. Prior Knowledge

2. Preparation

3. Modeling

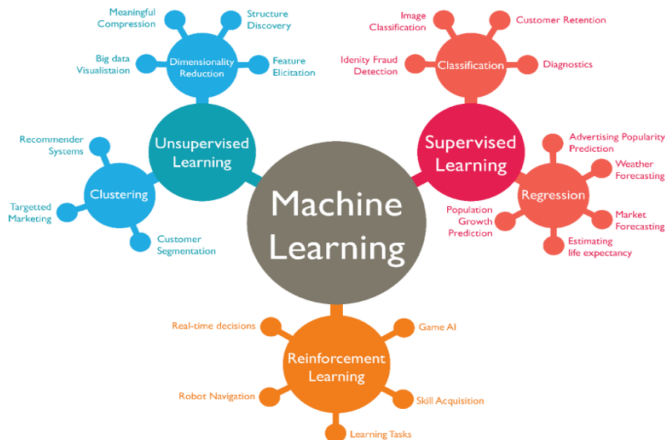
4. Application

5. Knowledge

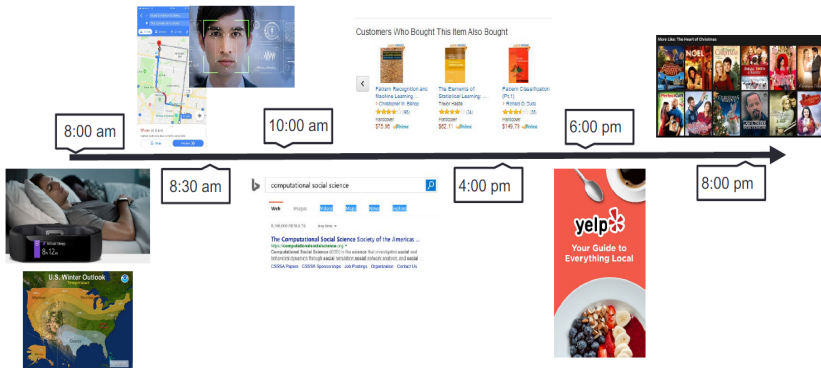
TOPIC 3 : MACHINE LEARNING

What is Machine Learning?

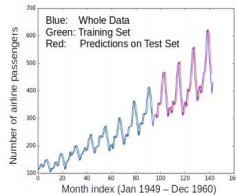
Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.



A day in our life with ML techniques...

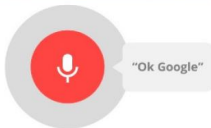
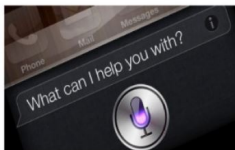


- Designing algorithms that **ingest data** and **learn a model** of the data.
- The learned model can be used to
 - 1 Detect **patterns/structures/themes/trends** etc. in the data
 - 2 Make **predictions** about future data and make decisions



- Modern ML algorithms are heavily **"data-driven"**.
- Optimize a performance criterion using example data or **experience**.

Broadly applicable in many domains (e.g., internet, robotics, healthcare and biology, computer vision, NLP, databases, computer systems, finance, etc.).

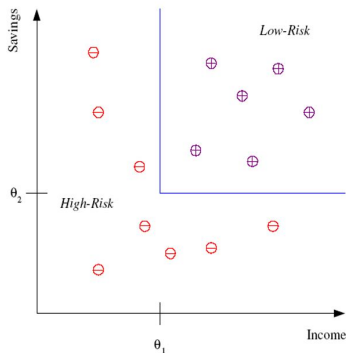


Predictive Policing

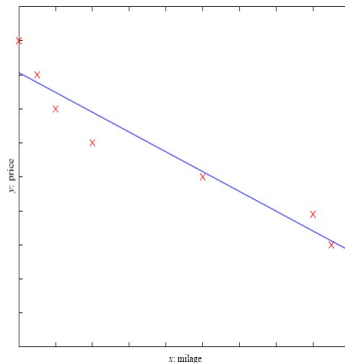


Online Fraud Detection

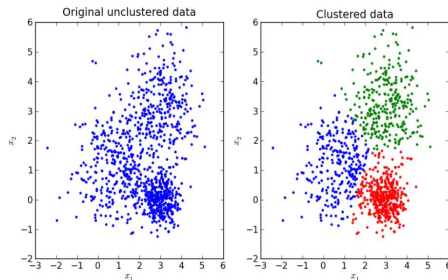
- **Example:** Credit scoring.
- Differentiating between low-risk and high-risk customers from their income and savings.
- **Discriminant:** IF $\text{Income} > \theta_1$ AND $\text{Savings} > \theta_2$ THEN low-risk ELSE high-risk.
- **Classification:** Learn a linear/nonlinear separator (the "model") using training data consisting of input-output pairs (each output is discrete-valued "label" of the corresponding input).
- Use it to predict the labels for new "test" inputs.
- **Other Applications:** Image Recognition, Spam Detection, Medical Diagnosis.



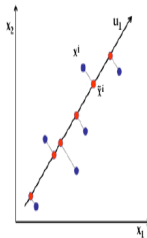
- **Example:** Price of a used car.
- X : car attributes; Y : price and $Y = f(X, \theta)$
- $f()$ is the model and θ is the model parameters.
- **Regression:** Learn a line/curve (the "model") using training data consisting of Input-output pairs (each output is a real-valued number).
- Use it to predict the outputs for new "test" inputs.
- **Other Applications:** Price Estimation, Process Improvement, Weather Forecasting.



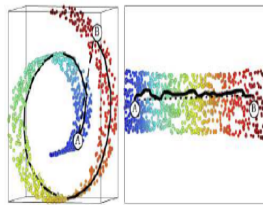
- **Given:** Training data in form of unlabeled instances $\{x_1, x_2, \dots, x_N\}$
- **Goal:** Learn the intrinsic latent structure that summarizes/explains data
- **Clustering:** Learn the grouping structure for a given set of unlabeled inputs.
- Homogeneous groups as latent structure: **Clustering**
- **Other Applications:** Topic Modelling, Image Segmentation, Social Networking.



- **Given:** Training data in form of **unlabeled instances** $\{x_1, x_2, \dots, x_N\}$
- **Dimensionality Reduction:** Learn a Low-dimensional representation for a given set of high-dimensional inputs
- **Note:** DR also comes in supervised flavors (supervised DR)
- **Other Applications:** facial recognition, computer vision and image compression.



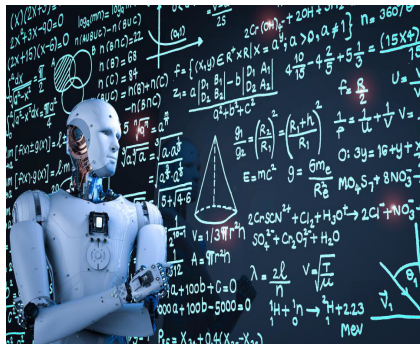
Two-dim to one-dim
linear projection



Three-dim to two-dim
nonlinear projection
(a.k.a. manifold learning)

TOPIC 4 : ARTIFICIAL INTELLIGENCE

- What is Artificial Intelligence?
- What are the main challenges?
- What are the applications of AI?
- What are the issues raised by AI?
- On September 1955, a project was proposed by McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon introducing formally for the first time the term "Artificial Intelligence".



- AI is about electronic device able to mimic human thinking:
 - 1 Artificial Intelligence.
 - 2 One famous class of AI algorithms are called neural networks.
 - 3 Android are close to humans in shape so they must think like human.
- Most AI algorithms do not aim at reproducing human reasoning.
- "The study and design of intelligent agents" where an intelligent agent is a system that perceives its environment and takes actions that maximize its chances of success - Frequent definition of AI.
- "In from three to eight years we will have a machine with the general intelligence of an average human being." - Marvin Minsky (1970, Life Magazine).

- Originates from 1920 (NY)
- First use of neural networks to control autonomous cars (1989)
- Four US states allow self-driving cars (2013)
- First known fatal accident (May 2016)
- Singapore launched the first self-driving taxi service (Aug. 2016)
- A Arizona pedestrian was killed by an Uber self-driving car (March 2018).



- Voice recognition tool "Harpy" masters about 1000 words (1970s, CMU, US Defense).
- System capable of analyzing entire word sequences (1980).
- Siri was the first modern digital virtual assistant installed on a smartphone (2011).
- Watson won the TV show Jeopardy! (2011).



TOPIC 5 : BIG DATA



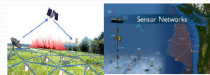
Social media and networks
(All of us are generating data)



Scientific instruments
(Collecting all sorts of data)



Mobile devices
(Tracking all objects all the time)



Sensor technology and networks
(Measuring all kinds of data)

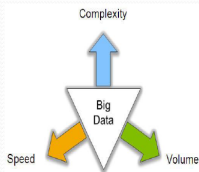
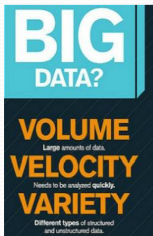
Now data is Big data!

"Big data is data whose scale, diversity, and complexity require new architecture, techniques, algorithms, and analytics to manage it and extract value and hidden knowledge from it" - Standard definition.

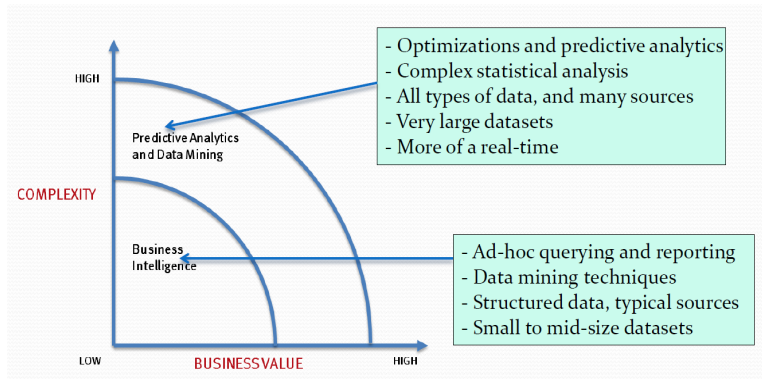
Difficulties related to (Big) data:

- The prediction must be accurate: difficult for some tasks like image classification, video captioning...
- The prediction must be quick: online recommendation should not take minutes.
- Data must be stored and accessible easily.
- It may be difficult to access all data at the same time. Data may come sequentially.

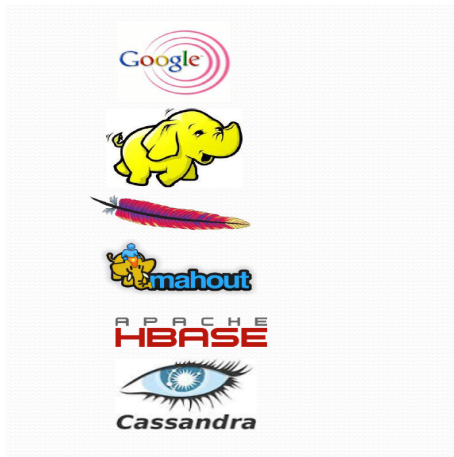




Big data is more real time in nature than traditional applications
(Massively parallel processing, scale out architectures are well suited for big data applications)...



- The Bottleneck is in technology: New architecture, algorithms, techniques are needed.
- Also in technical skills: Experts in using the new technology and dealing with Big data
- **Ethical issues:** Tay ("thinking about you") was an AI released by Microsoft via Twitter in 2016. It was shut down when the bot began to post in inflammatory and offensive tweets, only 16 hours after its launch.



Machine Learning:

- Theoretical ML (Explainable & Robust AI)
- Applied ML (Supervised & Unsupervised Problems)
- Deep Learning (Application to CV, IP, IOT, etc.)

Big Data & Analytics:

- Business Analytics (Quality, Reliability, OR Modelling)
- Data Assimilation
- Methods for Big Data
- High-Dimensional Statistics

Other Areas:

- Quantum Computing
- 3D Modelling, AR & VR
- Blockchain & Cryptography
- Robotics

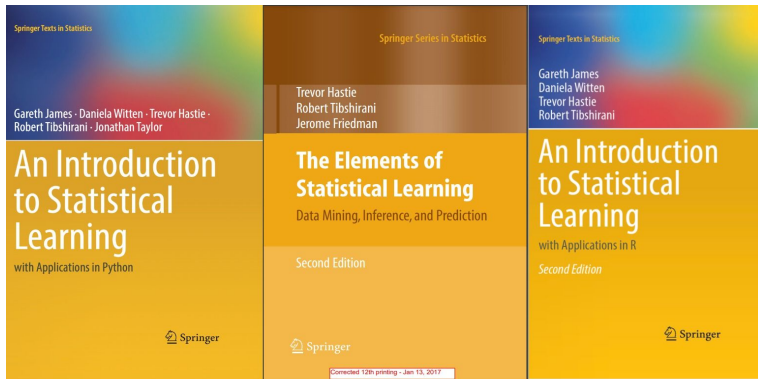
Shifting from Performance Driven to Risk Sensitive (Explainable AI)...



Human



- Emerging Research Areas in Statistics, Data Science & ML:
 - All the areas of Machine Learning.
 - High-Dimensional Small Sample Size Data.
 - Spatio-temporal Data.
 - Fairness in ML and Interpretable Deep Learning.
 - Collaborative research with other disciplines.
- Statistical Thinking:
 - Accepting randomness in formulating models and ideas.
 - Realizing and Analyzing the dependence of conclusions on assumptions.
 - Measuring uncertainty in some ways without forgetting dependence on assumptions.



What type of data are involved in the following applications?

- Weather forecasting
- Mobile usage of all customers of a service provider
- Anomaly (e.g. fraud) detection in a bank organization
- Person categorization, that is, identifying a human
- Air traffic control in an airport
- Streaming data from all flying air crafts of Boeing

Caution: "Prediction is very difficult, especially if it's about the future"
- Niels Bohr, Father of Quantum.

